

ASHABA JOSHUA JASPER

Data Scientist · AI / ML Engineer · Analytics & BI · Full-Stack Developer

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PROFESSIONAL SUMMARY

Data Scientist and AI/ML Engineer with a **First-Class Computer Science** background and a strong record of **shipping production data products end-to-end** - from data pipelines and LLM systems to observability dashboards and full-stack applications. Most recently delivered AI/ML, data-science, and analytics systems for a leading US radiology network (**Envision Radiology, via Reveloop Tech Systems**), spanning multi-LLM clinical analysis, document-intake automation, DICOM computer-vision deployment on Kubernetes, and Grafana telemetry dashboards across the Radiology Information System (RIS) and PACS ecosystem. Comfortable across the full stack of a modern DS/ML role: Python, SQL, TensorFlow/PyTorch, the OpenAI/Gemini/Anthropic model APIs, Azure (AKS, Application Insights, Key Vault), MLflow, and BI tooling (Grafana, Power BI, Tableau). Combines rigorous, metric-driven analysis with the engineering discipline to put models into production and keep them observable.

CORE COMPETENCIES

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| ■ Machine Learning & Deep Learning | ■ MLOps & Model Observability | ■ Statistical Analysis & Experimentation |
| ■ LLM / Generative AI Systems | ■ Full-Stack Development | ■ Data Storytelling & Reporting |
| ■ Predictive Modelling & Analytics | ■ Cloud & Kubernetes (Azure/AWS/GCP) | ■ Agile Project Delivery |
| ■ Data Pipeline & ETL Architecture | ■ Business Intelligence & Dashboards | ■ Cross-Functional Collaboration |
| ■ NLP & Computer Vision | ■ IT Systems & Network Administration | ■ Digital Transformation Strategy |

PROFESSIONAL EXPERIENCE

Data Scientist & AI/ML Engineer (Remote)

Reveloop Tech Systems (Ezrev) - Envision Radiology - Colorado Springs, USA February 2025 – June 2026

Delivered production AI/ML, data-science, and observability systems for a leading US radiology network across its Radiology Information System (RIS), DICOM imaging, and clinical document workflows.

- **Built RadCareLoop, a multi-LLM radiologist follow-up analyzer**, orchestrating a Gemini + GPT + Claude ensemble with a custom “Judge” arbitration layer; validated it against a 10,000-report gold-standard dataset at 97.9% accuracy and authored the full technical-architecture package.
- **Architected the Patient Document AI (Doc AI) intake pipeline** that automated processing of multi-page fax orders - splitting documents, filtering non-orders, and routing them through a 9-service orchestration layer (ICD, CPT, provider, facility, and priority mapping plus a sequential patient → insurance → encounter → exam matching chain) into SQL Server.
- **Engineered LLM-assisted CPT and ICD code-mapping services** that combined model inference with similarity matching to auto-map diagnosis and procedure codes, and resolved a critical patient-record duplication defect by hardening the entity-resolution logic in the patient-matching service.
- **Deployed DICOM computer-vision models on Azure Kubernetes Service (AKS)**, configuring node-pool autoscaling for imaging workloads and enforcing namespace resource quotas for FinOps cost control.
- **Stood up MLOps and observability across the stack** using MLflow for prompt-version and model-accuracy tracking and Azure Application Insights for end-to-end tracing of the Doc AI and DICOM SCP services.
- **Designed and shipped Grafana telemetry dashboards** for the RIS/PACS ecosystem (including the DICOM Worklist dashboard), implementing role-based access with Grafana Teams so units such as the PACS team

saw only their relevant metrics; presented the dashboards to stakeholders and defined the tracked-metric roadmap.

- **Owned the Python engineering foundation for these services** (REST APIs, pydantic data models, Azure Key Vault secrets management, and Postman validation suites), and delivered complete documentation, walkthroughs, and a clean handover across all four workstreams.

Lead Analyst & E-Commerce Developer

Excellent Shop - Kampala, Uganda *February 2024 – January 2025*

- **Architected and launched a full-stack e-commerce platform** integrating inventory tracking, secure payment gateways, and automated pricing, driving a 25% increase in online sales revenue.
- **Implemented real-time analytics dashboards** to monitor sales trends, customer behaviour, and conversion funnels, improving conversion rates by 15%.
- **Optimised SEO and site performance** for a 30% lift in organic traffic through structured-data markup, Core Web Vitals improvements, and content optimisation.
- **Automated stock-management workflows** with Python and SQL, cutting manual inventory errors by 40% and enabling real-time stock alerts.

Data Engineer & Digital Marketing Analyst

Uganda Bookshop - Kampala, Uganda *May 2023 – August 2024*

- **Designed automated ETL pipelines** using Python, SQL, and Apache Airflow to consolidate 5+ disparate sources, improving cross-departmental data accessibility by 50%.
- **Built ML models for customer segmentation and sales forecasting** with Scikit-learn and XGBoost, sharpening targeted marketing and contributing to a 40% rise in online sales.
- **Developed interactive BI dashboards** in Tableau and Power BI with live feeds, enabling real-time KPI monitoring that lifted customer engagement by 35%.

Mobile Application Developer

Centenary Publishing - Kampala, Uganda *August 2023 – April 2024*

- **Led the digitisation of print publications** into interactive e-books (Uganda Youth Praise, Come and Worship), increasing digital downloads by 10% in the first quarter.
- **Developed cross-platform mobile apps** with Flutter and Firebase using an offline-first architecture for low-connectivity regions across Uganda.

Data Analyst & Fleet Management Software Developer

Blue Pearls Company Ltd & Uganda Transporters - Nakawa, Uganda *April 2021 – June 2024*

- **Built and trained ML models** (Random Forests, XGBoost) on real-time IoT sensor data to predict maintenance needs, cutting unexpected breakdowns by 30% and upkeep costs by 20%.
- **Deployed an IoT-integrated fleet-management system** with geolocation tracking and predictive-maintenance scheduling, improving fleet uptime by 18%.
- **Designed real-time fleet analytics dashboards** for logistics tracking and route optimisation, enabling data-driven fleet operations.

Software Engineer Intern - Operations & Monitoring Division

MTN Uganda - Kampala, Uganda *July 2023 – September 2023*

- **Configured and maintained Grafana and the ELK Stack** for real-time infrastructure monitoring, strengthening system uptime and incident response.
- **Developed automation scripts** that reduced manual monitoring workloads by 25% across the network operations centre.
- **Optimised SQL queries and database performance**, improving analytics and reporting speed by 30% for monitoring dashboards.

Lead Technical & Business Analyst

Decades Investments - Kampala, Uganda *September 2019 – February 2021*

- **Developed and launched a full e-commerce store**, increasing online sales by 25% through optimised UI/UX and seamless checkout.
- **Designed an inventory-management system**, reducing manual stock-tracking errors by 20% and improving supply-chain visibility.

EDUCATION & ACADEMIC EXCELLENCE

BACHELOR OF COMPUTER SCIENCE

First Class Honours · GPA 4.62 / 5.0 · GDSC Lead

Uganda Christian University, Mukono, Uganda | January 2022 – July 2024

Academic Highlights

- **Graduated with First Class Honours (4.62/5.0)**, excelling across Data Science, Machine Learning, Software Engineering, and Algorithms coursework.
- **Strong mathematics and CS foundation:** Discrete Mathematics, Linear Algebra, Probability & Statistics, Calculus, Database Systems, Computer Networks, Software Architecture, and Distributed Systems.

Capstone Project

- **Help Anonymous App (270+ active users)** - a full-stack AI-powered mental-health platform in Flutter integrating anonymous counselling, NLP sentiment analysis (BERT/LSTM), and real-time chat; delivered end-to-end from requirements through deployment and user acquisition.

University Leadership & Impact

- **Google Developer Student Club (GDSC) Lead (2022–2023)** - led 100+ students through workshops on ML, Python, TensorFlow, and Cloud Computing, and organised hackathons and mentorship producing real-world project outcomes.
- **Ranked among Africa's Top 100** in the Google Developer Community for Machine Learning & Cloud - a continental recognition of technical leadership and impact.
- **General Secretary, Data Science Society** - ran workshops, mentorship, and collaborative research that raised student competency in Data Science and ML.

Junior Cybersecurity Analyst | I4G Scholarship Recipient - *Cisco Networking Academy, April 2023 – August 2023*

- **Competitively selected scholarship recipient** specialising in network security, threat analysis, penetration testing, and firewall management, with hands-on labs in vulnerability assessment and intrusion detection.

Certificate in Software Engineering with Python - *MTN ACE Program, Refactory Academy, 2023*

- **Industry-sponsored intensive** in Python, full-stack development, and production-grade application architecture, applying advanced data structures, algorithms, and API integration patterns.

Uganda Advanced Certificate of Education (UACE) - *Namityango College, Mukono, 2018 – 2019*

- **Studied Physics, Chemistry, Mathematics (PCM)** at one of Uganda's most prestigious institutions, and led the NASTECH Club to a Best Club Award (2019).

TECHNICAL PROFICIENCIES

Programming & Scripting: Python, SQL, R, JavaScript, TypeScript, C#/.NET, Dart, Bash, C++

Data Science & ML: Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, LightGBM, BERT, Hugging Face, OpenCV

Generative AI / LLMs: OpenAI (GPT), Google Gemini, Anthropic Claude APIs, LLM ensembles & evaluation, prompt engineering, RAG

Data Engineering & ETL: Apache Airflow, Apache Spark, Pandas, NumPy, SQLAlchemy, Dask, SQL Server, PostgreSQL, MySQL

BI & Visualisation: Grafana, Power BI, Tableau, Matplotlib, Seaborn, Plotly

Cloud & Infrastructure: Azure (AKS, Application Insights, Key Vault), AWS (SageMaker, S3, Lambda), GCP (BigQuery, Vertex AI)

MLOps & Observability: MLflow, DVC, Docker, Kubernetes, CI/CD, Application Insights, model & experiment tracking

Web & Mobile: Next.js, React, Django, Flask, FastAPI, Flutter, Firebase, REST APIs, GraphQL

Databases: SQL Server, PostgreSQL, MySQL, MongoDB, Redis, Supabase, Firebase

IT & Security: Linux administration, network configuration, firewall management, ELK Stack, ITIL practices

NOTABLE PROJECTS

RadCareLoop - Multi-LLM Radiology Analyzer (Envision Radiology)

- AI ensemble combining Gemini, GPT, and Claude with a Judge arbitration layer, validated at 97.9% accuracy against a 10,000-report gold-standard dataset for radiologist follow-up analysis.

Patient Document AI Pipeline (Envision Radiology)

- Automated multi-page fax order intake through a 9-service orchestration layer (code mapping, provider/facility matching, and a sequential patient–insurance–encounter–exam chain) writing to SQL Server, with App Insights tracing.

DICOM Vision on Kubernetes (Envision Radiology)

- Deployed computer-vision models for DICOM imaging on Azure AKS with autoscaling node pools and FinOps resource quotas, plus MLflow tracking for accuracy and prompt versions.

RIS/PACS Grafana Telemetry Suite (Envision Radiology)

- Grafana observability dashboards (incl. the DICOM Worklist dashboard) with team-based access control across the RIS ecosystem, presented to stakeholders to drive the metric roadmap.

AI-Powered Mental-Health Platform - Help Anonymous

- NLP sentiment-analysis engine (BERT/LSTM) detecting distress indicators, integrated into a Flutter app serving 270+ users with anonymous counselling.

Predictive Maintenance for Fleet Operations

- Random Forest and XGBoost models on real-time IoT sensor data, reducing unexpected breakdowns by 30% and operational costs by 20%.

AWARDS & HONOURS

- **Gold Award - Queen's Commonwealth Essay Competition (2021)**, international recognition for exceptional writing.
- **First Class Honours** - BSc Computer Science, Uganda Christian University (2024), GPA 4.62/5.0.
- **Top 100 in Africa** - Google Developer Community for Machine Learning & Cloud (2022–2023).
- **I4G Cybersecurity Scholarship** - Cisco Networking Academy (2023).

LICENSES & CERTIFICATIONS

- **IBM Data Science Professional Certificate** - Coursera (2026).
- **DeepLearning.AI - Machine Learning Specialization (Andrew Ng)** - Coursera (2026).
- **Junior Cybersecurity Analyst & Cyber Threat Management** - Cisco Networking Academy (2023).
- **Introduction to Machine Learning** - Kaggle (2023–2025).
- **Google Cloud Essentials** - Google Cloud Skills Boost (2023).
- **Digital Marketing Professional Certificate** - CertiProf (2023).

Full certification portfolio available upon request.

COMMUNITY LEADERSHIP & VOLUNTEERING

Founder & Program Director - Learnnovate

Technology Education Non-Profit - Uganda *December 2022 – Present*

- **Built and runs technology-education programs** reaching 200+ learners across multiple schools, focused on digital literacy and coding skills.

Data Analyst (Volunteer)

Radiant Smile Foundation - Uganda June 2023 – Present

- **Created impact-measurement dashboards** to analyse program effectiveness, enabling data-driven resource allocation and stronger grant reporting.

REFERENCES

Ms. Karibwije Ruth

General Manager, Uganda Bookshop

On request

| On request

Dr. Innocent Indibatya

Head of Department, Computing Sciences

Uganda Christian University

On request

| On request

Additional references available upon request. · Last updated: June 2026